

Problem 1. CLT with Two-Way Dependence. We follow the description of Menzel (2021) in Lecture note 17. Recall the representation of the RCE array $Z = (Z_{gh})_{1 \leq g, h < \infty}$:

$$Z_{gh} = f(\alpha, \beta_g, \gamma_h, \varepsilon_{gh}), \quad 1 \leq g, h < \infty,$$

where α , $(\beta_g)_{g=1}^\infty$, $(\gamma_h)_{h=1}^\infty$, and $(\varepsilon_{gh})_{1 \leq g, h < \infty}$ are IID $U[0, 1]$. We condition on α and drop it in what follows. We assume Z_{gh} is scalar with mean zero and finite variance. Recall

$$Z_{gh} = a_g + b_h + v_{gh} + e_{gh},$$

where

$$\begin{aligned} a_g &= \mathbb{E}[Z_{gh} \mid \beta_g], & v_{gh} &= \mathbb{E}[Z_{gh} \mid \beta_g, \gamma_h] - \mathbb{E}[Z_{gh} \mid \beta_g] - \mathbb{E}[Z_{gh} \mid \gamma_h], \\ b_h &= \mathbb{E}[Z_{gh} \mid \gamma_h], & e_{gh} &= Z_{gh} - \mathbb{E}[Z_{gh} \mid \beta_g, \gamma_h]. \end{aligned}$$

Assume as in note 17 that

$$v_{gh} = c_g d_h, \quad c_g = g(\beta_g), \quad d_h = d(\gamma_h).$$

Suppose we observe (Z_{gh}) for $g = 1, \dots, G$ and $h = 1, \dots, G$. Define the 5×1 vector

$$S_G = \begin{pmatrix} \sqrt{G} \bar{a}_G \\ \sqrt{G} \bar{b}_G \\ \sqrt{G} \bar{c}_G \\ \sqrt{G} \bar{d}_G \\ G \bar{e}_{GG} \end{pmatrix},$$

where $\bar{a}_G = \frac{1}{G} \sum_{g=1}^G a_g$, $\bar{b}_G$, $\bar{c}_G$, and $\bar{d}_G$ are defined analogously, and $\bar{e}_{GG} = \frac{1}{GG} \sum_{g=1}^G \sum_{h=1}^G e_{gh}$. You will show that

$$S_G \rightarrow_d N(0, \Sigma)$$

as $G \rightarrow \infty$, where the expression for Σ is given in note 17.

Consider the σ -algebras

$$\mathcal{F}_{G,g} = \begin{cases} \{\Omega, \emptyset\}, & g = 0, \\ \sigma(\beta_1, \dots, \beta_g, \gamma_1, \dots, \gamma_g, (\varepsilon_{g'1})_{g'=1}^G, \dots, (\varepsilon_{g'g})_{g'=1}^G), & g = 1, \dots, G. \end{cases}$$

(a) Calculate $\mathbb{E}[S_G \mid \mathcal{F}_{G,g}]$ for $g = 0, \dots, G$. Give an expression for the martingale differences $X_{G,g} = \mathbb{E}[S_G \mid \mathcal{F}_{G,g}] - \mathbb{E}[S_G \mid \mathcal{F}_{G,g-1}]$, $g = 1, \dots, G$.

(b) Show that

$$\max_{1 \leq g \leq G} \|X_{G,g}\| \rightarrow_p 0$$

as $G \rightarrow \infty$. State any additional assumptions you use.

(c) Show that

$$\sum_{g=1}^G (X_{G,g})(X_{G,g})' \rightarrow_p \Sigma$$

as $G \rightarrow \infty$. State any additional assumptions you use.

(d) Show that

$$\limsup_{G \rightarrow \infty} \mathbb{E} \left[\max_{1 \leq g \leq G} \|X_{G,g}\|^2 \right] < \infty.$$

State any additional assumptions you use.

In view of (b)–(d), it follows by a CLT for triangular arrays of martingale differences (e.g., Hall and Heyde, (1980, Theorem 3.2)) that $S_G \rightarrow_d N(0, \Sigma)$ as $G \rightarrow \infty$.

Solution. *Preliminaries.* Under the Aldous–Hoover representation, the random variables $\{\beta_g\}_{g \geq 1}$, $\{\gamma_h\}_{h \geq 1}$, $\{\varepsilon_{gh}\}_{g,h \geq 1}$ are mutually independent; a_g and c_g are $\sigma(\beta_g)$ -measurable with mean zero, b_h and d_h are $\sigma(\gamma_h)$ -measurable with mean zero, and the cell residual $e_{g'h}$ satisfies $\mathbb{E}[e_{g'h} \mid \beta_{g'}, \gamma_h] = 0$. Write $\nu(\beta, \gamma) := \mathbb{E}[Z_{11} \mid \beta_1 = \beta, \gamma_1 = \gamma]$ for the (β, γ) -conditional mean, so $e_{g'h} = Z_{g'h} - \nu(\beta_{g'}, \gamma_h)$. Define the β -marginalized residual

$$\tilde{e}(\gamma, \varepsilon) := \mathbb{E}[Z_{11} \mid \gamma_1 = \gamma, \varepsilon_{11} = \varepsilon] - b(\gamma),$$

so $\mathbb{E}[\tilde{e}(\gamma_h, \varepsilon_{g'h}) \mid \gamma_h] = 0$, $\mathbb{E}[\tilde{e}(\gamma_h, \varepsilon_{g'h})^2] < \infty$ (inherited from $\mathbb{E}[e^2] < \infty$). By construction $e_{g'h} - \tilde{e}(\gamma_h, \varepsilon_{g'h})$ has mean zero given $(\gamma_h, \varepsilon_{g'h})$, equivalently $\mathbb{E}[e_{g'h} \mid \gamma_h, \varepsilon_{g'h}] = \tilde{e}(\gamma_h, \varepsilon_{g'h})$.

(a) *Components 1–4.* Since a_ℓ is $\sigma(\beta_\ell)$ -measurable, $\mathbb{E}[a_\ell \mid \mathcal{F}_{G,g}] = a_\ell \mathbf{1}\{\ell \leq g\}$ (and equals $\mathbb{E}[a_\ell] = 0$ when $\ell > g$ by independence). Hence

$$\mathbb{E}[\sqrt{G} \bar{a}_G \mid \mathcal{F}_{G,g}] = \frac{1}{\sqrt{G}} \sum_{\ell=1}^g a_\ell,$$

and analogously for the other three components, replacing a_ℓ by b_ℓ , c_ℓ , and d_ℓ respectively:

$$\mathbb{E}[\sqrt{G} \bar{b}_G \mid \mathcal{F}_{G,g}] = \frac{1}{\sqrt{G}} \sum_{\ell=1}^g b_\ell, \quad \mathbb{E}[\sqrt{G} \bar{c}_G \mid \mathcal{F}_{G,g}] = \frac{1}{\sqrt{G}} \sum_{\ell=1}^g c_\ell,$$

$$\mathbb{E}[\sqrt{G} \bar{d}_G \mid \mathcal{F}_{G,g}] = \frac{1}{\sqrt{G}} \sum_{\ell=1}^g d_\ell.$$

Component 5. Compute $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}]$ by casework on (g', h) :

- (i) $g' \leq g, h \leq g$: all of $\beta_{g'}, \gamma_h, \varepsilon_{g'h}$ lie in $\mathcal{F}_{G,g}$, so $e_{g'h}$ itself is $\mathcal{F}_{G,g}$ -measurable and $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = e_{g'h}$.
- (ii) $g' \leq g, h > g$: $\beta_{g'}$ is in $\mathcal{F}_{G,g}$; γ_h and $\varepsilon_{g'h}$ are independent of $\mathcal{F}_{G,g}$. So $\mathbb{E}[Z_{g'h} \mid \mathcal{F}_{G,g}] = \mathbb{E}[Z_{g'h} \mid \beta_{g'}] = a_{g'}$ and $\mathbb{E}[\nu(\beta_{g'}, \gamma_h) \mid \mathcal{F}_{G,g}] = \mathbb{E}[\nu(\beta_{g'}, \gamma_h) \mid \beta_{g'}] = a_{g'}$, so $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = 0$.
- (iii) $g' > g, h \leq g$: γ_h and $\varepsilon_{g'h}$ are in $\mathcal{F}_{G,g}$; $\beta_{g'}$ is independent of $\mathcal{F}_{G,g}$. Hence $\mathbb{E}[Z_{g'h} \mid \mathcal{F}_{G,g}] = \mathbb{E}[Z_{g'h} \mid \gamma_h, \varepsilon_{g'h}] = b_h + \tilde{e}(\gamma_h, \varepsilon_{g'h})$ and $\mathbb{E}[\nu(\beta_{g'}, \gamma_h) \mid \mathcal{F}_{G,g}] = b_h$, so $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = \tilde{e}(\gamma_h, \varepsilon_{g'h})$.
- (iv) $g' > g, h > g$: all latent coordinates are independent of $\mathcal{F}_{G,g}$, so $\mathbb{E}[Z_{g'h} \mid \mathcal{F}_{G,g}] = 0$ and $\mathbb{E}[\nu(\beta_{g'}, \gamma_h) \mid \mathcal{F}_{G,g}] = 0$, giving $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = 0$.

Summing,

$$\mathbb{E}[G \bar{e}_{GG} \mid \mathcal{F}_{G,g}] = \frac{1}{G} \sum_{g' \leq g} \sum_{h \leq g} e_{g'h} + \frac{1}{G} \sum_{g' > g} \sum_{h \leq g} \tilde{e}(\gamma_h, \varepsilon_{g'h}).$$

Martingale differences. Collecting,

$$X_{G,g} = \frac{1}{\sqrt{G}}(a_g, b_g, c_g, d_g, 0)' + (0, 0, 0, 0, \Delta_g)',$$

where taking the difference of $\mathbb{E}[G \bar{e}_{GG} \mid \mathcal{F}_{G,g}]$ at steps g and $g-1$ yields

$$\Delta_g = \frac{1}{G} \left[\underbrace{\sum_{g'=1}^g e_{g'g}}_{\text{column } g, \text{ fully revealed}} + \underbrace{\sum_{g'=g+1}^G \tilde{e}(\gamma_g, \varepsilon_{g'g})}_{\text{column } g, \text{ projected}} + \underbrace{\sum_{h=1}^{g-1} (e_{gh} - \tilde{e}(\gamma_h, \varepsilon_{gh}))}_{\text{row } g \text{ cells, projection upgraded}} \right].$$

The three sums correspond to the cells whose status changes at step g : column g cells become fully measurable (Case (i)) or move from Case (iv) to Case (iii); row g cells with $h < g$ move from Case (iii) to Case (i), replacing $\tilde{e}(\gamma_h, \varepsilon_{gh})$ by e_{gh} . Straightforward verification using $\mathbb{E}[a_g] = \mathbb{E}[b_g] = \mathbb{E}[c_g] = \mathbb{E}[d_g] = 0$ and $\mathbb{E}[e_{gh} - \tilde{e}(\gamma_h, \varepsilon_{gh}) \mid \gamma_h, \varepsilon_{gh}] = 0$ gives $\mathbb{E}[X_{G,g} \mid \mathcal{F}_{G,g-1}] = 0$.

(b) For components 1–4, fix $\epsilon > 0$:

$$\mathbb{P}\left(\max_{g \leq G} |a_g| > \epsilon \sqrt{G}\right) \leq G \mathbb{P}(|a_1| > \epsilon \sqrt{G}) \leq \frac{\mathbb{E}[a_1^2 \mathbf{1}\{|a_1| > \epsilon \sqrt{G}\}]}{\epsilon^2},$$

using $\mathbb{P}(|a_1| > \epsilon \sqrt{G}) \leq \mathbb{E}[a_1^2 \mathbf{1}\{|a_1| > \epsilon \sqrt{G}\}]/(\epsilon^2 G)$. The right-hand side vanishes by dominated convergence as $\mathbf{1}\{|a_1| > \epsilon \sqrt{G}\} \rightarrow 0$ a.s. and a_1^2 is integrable. The same argument applies to b_g, c_g, d_g .

For component 5, write $\Delta_g = G^{-1}(S_{1,g} + S_{2,g} + S_{3,g})$ with

$$S_{1,g} = \sum_{g'=1}^g e_{g'g}, \quad S_{2,g} = \sum_{g'=g+1}^G \tilde{e}(\gamma_g, \varepsilon_{g'g}), \quad S_{3,g} = \sum_{h=1}^{g-1} (e_{gh} - \tilde{e}(\gamma_h, \varepsilon_{gh})).$$

Conditional independence. Conditional on γ_g , $(\beta_{g'}, \varepsilon_{g'g})_{g'=1, \dots, g}$ are iid and jointly independent of γ_g (by mutual independence of Aldous–Hoover latents); hence $\{e_{g'g}\}_{g'=1}^g$ are conditionally iid given γ_g with $\mathbb{E}[e_{g'g} \mid \gamma_g] = \mathbb{E}[\mathbb{E}[e_{g'g} \mid \beta_{g'}, \gamma_g] \mid \gamma_g] = 0$. Identical logic applies to $S_{2,g}$ (summands $\tilde{e}(\gamma_g, \varepsilon_{g'g})$ conditionally iid across $g' > g$ given γ_g) and to $S_{3,g}$ (summands $e_{gh} - \tilde{e}(\gamma_h, \varepsilon_{gh})$ conditionally iid across $h < g$ given β_g , with conditional mean zero by construction of $\tilde{e}$).

Rosenthal bound. For any $p \in (2, 2 + \delta]$,

$$\mathbb{E}[|S_{1,g}|^p \mid \gamma_g] \leq C_p \{g^{p/2} (\mathbb{E}[e_{11}^2 \mid \gamma_g])^{p/2} + g \mathbb{E}[|e_{11}|^p \mid \gamma_g]\} = O_{a.s.}(g^{p/2}).$$

Taking outer expectation, $\mathbb{E}|S_{1,g}|^p \leq C g^{p/2}$ (uniformly in $g \leq G$). Analogously, $\mathbb{E}|S_{2,g}|^p \leq C (G - g)^{p/2}$ and $\mathbb{E}|S_{3,g}|^p \leq C g^{p/2}$. Since $\max(g, G - g)^{p/2} \leq G^{p/2}$,

$$\mathbb{E}|\Delta_g|^p \leq G^{-p} \cdot 3^{p-1} (\mathbb{E}|S_{1,g}|^p + \mathbb{E}|S_{2,g}|^p + \mathbb{E}|S_{3,g}|^p) \leq C' G^{-p/2}$$

uniformly in $1 \leq g \leq G$. Therefore

$$\mathbb{P}\left(\max_{g \leq G} |\Delta_g| > \epsilon\right) \leq G \cdot \frac{C' G^{-p/2}}{\epsilon^p} = \frac{O(G^{1-p/2})}{\epsilon^p} \rightarrow 0$$

for any $p > 2$. Combining, $\max_g \|X_{G,g}\| \rightarrow_p 0$.

Assumptions used: $\mathbb{E}[a_1^2], \mathbb{E}[b_1^2], \mathbb{E}[c_1^2], \mathbb{E}[d_1^2] < \infty$ (for components 1–4, via dominated convergence); $\mathbb{E}[e_{11}]^{2+\delta} < \infty$ for some $\delta > 0$ (for component 5, via Rosenthal).

(c) Decompose $\sum_g X_{G,g} X'_{G,g} = A_G + B_G + C_G$, where

$$A_G = \frac{1}{G} \sum_{g=1}^G w_g w'_g \text{ with } w_g = (a_g, b_g, c_g, d_g)', \text{ extended by a zero 5}^{\text{th}} \text{ row,}$$

$$B_G = \sum_{g=1}^G \Delta_g^2 \text{ in the } (5, 5) \text{ entry, zero elsewhere,}$$

$$C_G = \sum_{g=1}^G \frac{1}{\sqrt{G}} (a_g, b_g, c_g, d_g)' \Delta_g \quad (\text{rank-one cross-terms}).$$

$A_G \rightarrow_p \mathbb{E}[w_1 w'_1]$ by the LLN for iid sequences with finite second moments. The limit is a 4×4 block Σ_w whose entries are derived as follows: a_1, c_1 are $\sigma(\beta_1)$ -measurable and b_1, d_1 are $\sigma(\gamma_1)$ -measurable, with $\beta_1 \perp\!\!\!\perp \gamma_1$ and all four marginals mean zero. Hence

$$\mathbb{E}[a_1 b_1] = \mathbb{E}[a_1] \mathbb{E}[b_1] = 0, \quad \mathbb{E}[a_1 d_1] = \mathbb{E}[b_1 c_1] = \mathbb{E}[c_1 d_1] = 0,$$

while $\mathbb{E}[a_1 c_1]$ and $\mathbb{E}[b_1 d_1]$ are generally nonzero (both terms are functions of the same latent). Diagonals are $\sigma_a^2, \sigma_b^2, \sigma_c^2, \sigma_d^2$, giving the block Σ_w with two non-zero off-diagonal pairs $\mathbb{E}[a_1 c_1]$ and $\mathbb{E}[b_1 d_1]$, in agreement with Lecture 17.

B_G convergence and cross- g orthogonality. The claim is $B_G \rightarrow_p \sigma_e^2 := \mathbb{E}[e_{11}^2]$. Recall $\Delta_g = G^{-1}(S_{1,g} + S_{2,g} + S_{3,g})$ from (b). The conditional expectations of step (a) telescope across $g = 0, \dots, G$ to $\mathbb{E}[G\bar{e}_{GG} \mid \mathcal{F}_{G,G}] - \mathbb{E}[G\bar{e}_{GG} \mid \mathcal{F}_{G,0}] = G\bar{e}_{GG}$, so $\sum_{g=1}^G \Delta_g = G\bar{e}_{GG}$. Since $\{\Delta_g\}_{g=1}^G$ are martingale differences (by the MD property of $X_{G,g}$), MD orthogonality gives $\sum_g \mathbb{E}[\Delta_g^2] = \text{Var}(\sum_g \Delta_g) = \text{Var}(G\bar{e}_{GG})$:

$$\text{Var}(G\bar{e}_{GG}) = G^{-2} \sum_{g', h, g'', h'} \text{Cov}(e_{g'h}, e_{g''h'}) = G^{-2} \sum_{g', h} \mathbb{E}[e_{g'h}^2] = \mathbb{E}[e_{11}^2] = \sigma_e^2,$$

using pairwise orthogonality of distinct cell residuals: for $(g', h) \neq (g'', h')$, $\mathbb{E}[e_{g'h} e_{g''h'}] = 0$ by $\mathbb{E}[e \mid \beta, \gamma] = 0$ (condition on the full (β, γ) -vector generating both cells and factor the expectation). For convergence in probability, Chebyshev plus a second-moment bound on $B_G - \mathbb{E}[B_G]$ suffices: by the Rosenthal bound from (b), $\text{Var}(\Delta_g^2) = O(G^{-2})$, so $\text{Var}(B_G) \leq G \cdot O(G^{-2}) = O(G^{-1}) \rightarrow 0$.

Cross-term C_G . The claim is $C_G \rightarrow_p 0$. For each g , (w_g, Δ_g) satisfy

$$\mathbb{E}[w_g \Delta_g \mid \mathcal{F}_{G,g-1}] = w_g \cdot \mathbb{E}[\Delta_g \mid \mathcal{F}_{G,g-1}] = 0$$

because $w_g = (a_g, b_g, c_g, d_g)'$ is $\sigma(\beta_g, \gamma_g)$ -measurable hence independent of $\mathcal{F}_{G,g-1}$, and Δ_g has mean zero conditional on $\mathcal{F}_{G,g-1}$; joint conditional mean zero follows from the tower property combined with independence of w_g from the cells entering Δ_g given $\mathcal{F}_{G,g-1}$. Hence $\{w_g \Delta_g / \sqrt{G}\}_{g=1}^G$ is a martingale-difference sequence, and by orthogonality

$$\mathbb{E} \left\| \sum_g \frac{w_g \Delta_g}{\sqrt{G}} \right\|^2 = \frac{1}{G} \sum_g \mathbb{E}[\|w_g\|^2 \Delta_g^2] \leq \frac{1}{G} \sum_g (\mathbb{E}\|w_g\|^4)^{1/2} (\mathbb{E}\Delta_g^4)^{1/2}.$$

Under $\mathbb{E}\|w_1\|^4 < \infty$ and the Rosenthal bound $\mathbb{E}\Delta_g^4 = O(G^{-2})$ (taking $p = 4$ in (b), requiring $\delta \geq 2$, i.e. $\mathbb{E}|e_{11}|^4 < \infty$), the sum is $O(G^{-1}) \rightarrow 0$, so $C_G \rightarrow_p 0$ by Markov.

Assembling: $\sum_g X_{G,g} X'_{G,g} \rightarrow_p \Sigma$ where Σ has the 4×4 block Σ_w in its upper-left 4×4 block and σ_e^2 in its $(5, 5)$ entry, all other entries zero — precisely the Σ of Lecture 17.

Assumptions used: $\mathbb{E}\|w_1\|^4 < \infty$, i.e. $\mathbb{E}|a_1|^4, \mathbb{E}|b_1|^4, \mathbb{E}|c_1|^4, \mathbb{E}|d_1|^4 < \infty$; $\mathbb{E}|e_{11}|^4 < \infty$ (for the Rosenthal bound with $p = 4$). These are implied by $\mathbb{E}|Z_{11}|^4 < \infty$.

(d) For components 1–4, using $\max_g a_g^2 \leq \sum_g a_g^2$,

$$\mathbb{E} \left[\max_{g \leq G} \frac{a_g^2}{G} \right] \leq \frac{1}{G} \sum_{g=1}^G \mathbb{E}[a_g^2] = \mathbb{E}[a_1^2] < \infty,$$

uniformly in G , and likewise for b_g^2, c_g^2, d_g^2 . For component 5, by Jensen's inequality and the Rosenthal bound from (b) with $p = 2 + \delta$:

$$\begin{aligned} \mathbb{E} \left[\max_{g \leq G} \Delta_g^2 \right] &\leq \left(\mathbb{E} \left[\max_{g \leq G} |\Delta_g|^{2+\delta} \right] \right)^{2/(2+\delta)} \leq \left(\sum_{g=1}^G \mathbb{E} |\Delta_g|^{2+\delta} \right)^{2/(2+\delta)} \\ &\leq (G \cdot O(G^{-(2+\delta)/2}))^{2/(2+\delta)} = O(G^{-\delta/(2+\delta)}), \end{aligned}$$

which is bounded (indeed vanishing). Summing the five component bounds gives $\limsup_{G \rightarrow \infty} \mathbb{E}[\max_g \|X_{G,g}\|^2] \leq \mathbb{E}[a_1^2] + \mathbb{E}[b_1^2] + \mathbb{E}[c_1^2] + \mathbb{E}[d_1^2] + 0 < \infty$.

Assumptions used: finite second moments for a, b, c, d ; $\mathbb{E}|e_{11}|^{2+\delta} < \infty$ for some $\delta > 0$ (for the max-over- g bound on Δ_g^2 via the Rosenthal argument of (b)).

Parts (b)–(d) verify the three conditions of the martingale CLT for triangular arrays of Hall and Heyde (1980, Theorem 3.2) — the martingale-difference version of the stable-convergence CLT stated as Theorem 1 of Lecture 10 (cf. Remark 1 therein, extending to triangular arrays). Hence $S_G \rightarrow_d N(0, \Sigma)$ as $G \rightarrow \infty$. $\square$

Problem 2. Narrative identification. Suppose you wish to estimate the parameters A and B in the model

$$Y_t' A = X_t' B + u_t',$$

where Y_t is $m \times 1$, A is $m \times m$ and invertible, X_t is $p \times 1$, B is $p \times m$, and u_t is $m \times 1$, where $p \geq m$. You observe an IID sample (X_t, Y_t) for $t = 1, \dots, n$.¹ We treat m as fixed and

¹Typically this is done in an SVAR context so there is time-series dependence, but that's the least of our worries.

let $n \rightarrow \infty$. The errors u_t are assumed to satisfy $\mathbb{E}[X_t u_t'] = 0$ and $\mathbb{E}[u_t u_t'] = I$, the $m \times m$ identity matrix. You may also assume $\mathbb{E}[X_t X_t']$ is finite and positive definite.

Rewrite the model as

$$Y_t' = X_t' B A^{-1} + u_t' A^{-1} = X_t' C + v_t',$$

where $C = B A^{-1}$ and $v_t = (A^{-1})' u_t$ has variance $\Sigma = (A^{-1})' A^{-1}$. Evidently, there is an identification problem: we can identify C from the moment $\mathbb{E}[X_t Y_t']$ and Σ from the moment $\mathbb{E}[v_t v_t']$, but we cannot disentangle A and B from C and Σ . Let $h(\Sigma)$ denote the lower-triangular Cholesky factor of Σ . That is, the unique lower-triangular matrix with positive diagonal elements for which $h(\Sigma)h(\Sigma)' = \Sigma$. For any rotation matrix Q (i.e., a square matrix for which $Q Q' = Q' Q = I$), we also have $h(\Sigma)Q(h(\Sigma)Q)' = h(\Sigma)Q Q' h(\Sigma)' = \Sigma$. Letting $\mathcal{Q}$ denote the set of all such Q , the identified set for A is $\{(h(\Sigma)')^{-1}Q : Q \in \mathcal{Q}\}$ and the identified set for B is $\{C(h(\Sigma)')^{-1}Q : Q \in \mathcal{Q}\}$.

Several authors (see, e.g., Antolín-Díaz and Rubio-Ramírez (AER, 2018); Ludvigson, Ma, and Ng (AEJ:Macro, 2021); Giacomini, Kitagawa, and Read (WP, 2025)) have advocated using “narrative restrictions”. In effect, the restrictions constrain the realizations of u_t on certain dates, which shrinks the identified set of A and B . Note that $u_t' = v_t' A = v_t' (h(\Sigma)')^{-1}Q$ for some $Q \in \mathcal{Q}$. Restricting u_t' on certain dates t allows us, with enough restrictions, to pin down a unique $Q \in \mathcal{Q}$, and thus “identify” the parameters A and B . For simplicity, let us assume that it is known that

$$R = \begin{pmatrix} u_1' \\ \vdots \\ u_m' \end{pmatrix},$$

where R is $m \times m$. Then

$$R = \underbrace{\begin{pmatrix} v_1' \\ \vdots \\ v_m' \end{pmatrix}}_{=V} (h(\Sigma)')^{-1} Q,$$

and so, assuming invertibility of V , we have

$$Q = h(\Sigma)' V^{-1} R.$$

Suppose you estimate C by regressing Y_t on X_t :

$$\hat{C} = \left(\frac{1}{n} \sum_{t=1}^n X_t X_t' \right)^{-1} \left(\frac{1}{n} \sum_{t=1}^n X_t Y_t' \right),$$

and estimate Σ from the OLS residuals:

$$\hat{\Sigma} = \frac{1}{n} \sum_{t=1}^n \hat{v}_t \hat{v}_t',$$

where $\hat{v}_t' = Y_t' - X_t' \hat{C}$. For any $Q \in \mathcal{Q}$, you can then estimate B using

$$\hat{C}(h(\hat{\Sigma})')^{-1}Q.$$

For the specific Q corresponding to the narrative restriction,

$$\begin{aligned}\hat{B} &= \hat{C}(h(\hat{\Sigma})')^{-1}h(\hat{\Sigma})'\hat{V}^{-1}R \\ &= \hat{C}\hat{V}^{-1}R,\end{aligned}$$

where

$$\hat{V} = \begin{pmatrix} \hat{v}_1' \\ \vdots \\ \hat{v}_m' \end{pmatrix}.$$

Derive the asymptotic distribution of $\hat{B}$ and use this, if possible, to develop a suitable inference procedure. State any additional assumptions you use.

Solution. Write $\Delta C = \hat{C} - C$ ($p \times m$) and $\Delta V = \hat{V} - V$ ($m \times m$). Throughout this problem, Σ denotes the $m \times m$ reduced-form variance $\mathbb{E}[v_t v_t'] = (A^{-1})' A^{-1}$ (unrelated to the 5×5 symbol Σ of Problem 1). The argument proceeds in six steps.

1. *Asymptotic expansion of $\hat{C}$.* Under iid sampling with $\mathbb{E}\|X_t\|^{2+\delta} < \infty$, $\mathbb{E}\|u_t\|^{2+\delta} < \infty$ for some $\delta > 0$, conditional homoskedasticity $\mathbb{E}[v_t v_t' | X_t] = \Sigma$, and $Q_{xx} := \mathbb{E}[X_t X_t']$ invertible, the standard OLS CLT yields

$$\sqrt{n} \text{vec}(\Delta C) \rightarrow_d \xi \sim N(0, \Omega_C), \quad \Omega_C = \Sigma \otimes Q_{xx}^{-1},$$

where $\Sigma = \mathbb{E}[v_t v_t']$. This follows from $\text{vec}(\Delta C) = (I_m \otimes (X'X/n)^{-1}) \cdot n^{-1/2} \sum_t \text{vec}(X_t v_t')$ with $\text{vec}(X_t v_t') = v_t \otimes X_t$, and Lindeberg–Lévy applied to the iid zero-mean summands $v_t \otimes X_t$ under the stated moment conditions ($\text{Var}(v_t \otimes X_t) = \Sigma \otimes Q_{xx}$).

2. *ΔV in terms of ΔC .* $\hat{v}_t = Y_t - \hat{C}'X_t = v_t - (\hat{C}' - C')X_t$, so $\hat{v}_t - v_t = -\Delta C'X_t$ and, stacking $t = 1, \dots, m$,

$$\hat{V} - V = -\tilde{X}'\Delta C, \quad \tilde{X} := [X_1 | X_2 | \dots | X_m] \in \mathbb{R}^{p \times m}.$$

3. *Asymptotic independence.* Decompose

$$n^{-1/2} \sum_{t=1}^n X_t v_t' = n^{-1/2} \sum_{t=1}^m X_t v_t' + n^{-1/2} \sum_{t=m+1}^n X_t v_t'.$$

The first piece is $O_p(1/\sqrt{n}) = o_p(1)$ since m is fixed, and the second piece is independent of $(X_1, v_1, \dots, X_m, v_m)$. Hence $\sqrt{n} \text{vec}(\Delta C)$ is asymptotically independent of $\mathcal{G} := \sigma(X_1, v_1, \dots, X_m, v_m)$; in particular, the limiting ξ in step 1 can be taken independent of $\mathcal{G}$.

4. *Linearization of $\hat{B}$.* Since V is invertible by assumption and $\hat{V} \rightarrow_p V$ (immediate from $\Delta V = -\tilde{X}'\Delta C = O_p(1/\sqrt{n})$), the smallest singular value satisfies $\sigma_{\min}(\hat{V}) \geq \sigma_{\min}(V) - \|\Delta V\| \geq \sigma_{\min}(V)/2 > 0$ on the event $\{\|\Delta V\| \leq \sigma_{\min}(V)/2\}$, whose probability tends to one. On this event, $\hat{V}$ is invertible and $\hat{V}^{-1} = V^{-1} - V^{-1}(\Delta V)V^{-1} + O_p(\|\Delta V\|^2) = V^{-1} - V^{-1}(\Delta V)V^{-1} + o_p(1/\sqrt{n})$. Thus,

$$\begin{aligned}\hat{B} - B &= (C + \Delta C)(V^{-1} - V^{-1}(\Delta V)V^{-1})R - CV^{-1}R \\ &= (\Delta C)V^{-1}R - CV^{-1}(\Delta V)V^{-1}R + o_p(1/\sqrt{n}).\end{aligned}$$

Substituting $\Delta V = -\tilde{X}'\Delta C$ from step 2,

$$\hat{B} - B = (\Delta C)V^{-1}R + CV^{-1}\tilde{X}'(\Delta C)V^{-1}R + o_p(1/\sqrt{n}) = M_n \cdot \Delta C \cdot N + o_p(1/\sqrt{n}),$$

where $M_n := I_p + CV^{-1}\tilde{X}'$ (a $p \times p$ random matrix, $\mathcal{G}$ -measurable) and $N := V^{-1}R$ (an $m \times m$ $\mathcal{G}$ -measurable matrix).

5. *Vectorized limit and stable convergence.* Using $\text{vec}(M_n(\Delta C)N) = (N' \otimes M_n) \text{vec}(\Delta C)$,

$$\sqrt{n} \text{vec}(\hat{B} - B) = K_n \cdot \sqrt{n} \text{vec}(\Delta C) + o_p(1), \quad K_n := N' \otimes M_n.$$

Since K_n is $\mathcal{G}$ -measurable and $\sqrt{n} \text{vec}(\Delta C) \rightarrow_d \xi$ $\mathcal{G}$ -stably by step 3 (with ξ independent of $\mathcal{G}$), the continuous-mapping/Slutsky variant for stable convergence (Lecture 10; see also the argument in PS3 Problem 1) gives

$$\sqrt{n} \text{vec}(\hat{B} - B) \rightarrow_d K_n \xi \quad (\mathcal{G}\text{-stably}).$$

The limit is a *mixed normal*: conditional on $\mathcal{G}$, $N(0, \Omega_B)$ with $\Omega_B := K_n \Omega_C K_n'$; unconditionally, a $\mathcal{G}$ -mixture.

Structural simplification. The conditional variance collapses cleanly. With $\Omega_C = \Sigma \otimes Q_{xx}^{-1}$ and $K_n = N' \otimes M_n$ (where $N = V^{-1}R$, $M_n = I_p + CV^{-1}\tilde{X}'$, both $\mathcal{G}$ -measurable),

$$\Omega_B = (N' \otimes M_n)(\Sigma \otimes Q_{xx}^{-1})(N \otimes M_n') = (N'\Sigma N) \otimes (M_n Q_{xx}^{-1} M_n').$$

The narrative-identification formula $V^{-1}R = A$ gives $N = A$. The structural normalization $\mathbb{E}[u_t u_t'] = I_m$ implies $\Sigma = (A^{-1})'(A^{-1}) = (AA')^{-1}$, hence

$$N'\Sigma N = A'(AA')^{-1}A = I_m.$$

Therefore

$$\Omega_B = I_m \otimes (M_n Q_{xx}^{-1} M_n').$$

The asymptotic variance is block-diagonal with m identical $p \times p$ blocks: each column of $\hat{B}$ has covariance $M_n Q_{xx}^{-1} M_n' / n$, and columns are asymptotically uncorrelated. Notably, Ω_B does not depend on the unobserved Σ or on A directly — a feature that makes inference particularly clean.

6. *Inference.* Plug in $\hat{M}_n := I_p + \hat{C}\hat{V}^{-1}\tilde{X}'$ and $\hat{Q}_{xx} := n^{-1} \sum_t X_t X_t'$:

$$\hat{\Omega}_B := I_m \otimes (\hat{M}_n \hat{Q}_{xx}^{-1} \hat{M}_n').$$

Consistency. $\hat{C} \rightarrow_p C$ and $\hat{V} \rightarrow_p V$ from steps 1–2 imply $\hat{M}_n \rightarrow_p M_n$; $\hat{Q}_{xx} \rightarrow_p Q_{xx}$ by the LLN. Hence $\hat{\Omega}_B \rightarrow_p \Omega_B$.

Multivariate t -statistic. For any non-random contrast $a \in \mathbb{R}^{pm}$, applying the continuous mapping theorem for stable convergence (to the smooth map $g(K, \xi, W) = a' K \xi / \sqrt{a' W a}$ on the positive-variance event)

$$\frac{a' \text{vec}(\hat{B} - B)}{\sqrt{a' \hat{\Omega}_B a / n}} \rightarrow_d N(0, 1)$$

on the event $\{a' \Omega_B a > 0\}$. This is the multivariate analogue of the t -statistic result of PS3 Problem 1 (cf. Lecture 10): stable convergence of a sequence with a consistent positive-definite variance estimator yields a standard-normal t -statistic. A standard $(1 - \alpha)$ confidence interval for $a' \text{vec}(B)$:

$$a' \text{vec}(\hat{B}) \pm z_{1-\alpha/2} \sqrt{a' \hat{\Omega}_B a / n}.$$

For an individual entry B_{ij} (the i -th coefficient on the j -th shock), the standard error simplifies to $\sqrt{[\hat{M}_n \hat{Q}_{xx}^{-1} \hat{M}_n']_{ii} / n}$, since the (j, j) block of $\hat{\Omega}_B$ is $\hat{M}_n \hat{Q}_{xx}^{-1} \hat{M}_n'$ regardless of j . Joint inference on multiple linear contrasts follows by the continuous mapping theorem applied to a Wald statistic:

$$T_n = n(\Lambda \text{vec}(\hat{B}) - r_0)' (\Lambda \hat{\Omega}_B \Lambda')^{-1} (\Lambda \text{vec}(\hat{B}) - r_0) \xrightarrow{d} \chi_{\text{rank}(\Lambda)}^2$$

under $\Lambda \text{vec}(B) = r_0$, on the event $\{\Lambda \Omega_B \Lambda' \succ 0\}$.

Assumptions used: iid (X_t, Y_t) with m fixed, $n \rightarrow \infty$; $\mathbb{E}\|X_t\|^{2+\delta}, \mathbb{E}\|u_t\|^{2+\delta} < \infty$ for some $\delta > 0$ (Lindeberg for $\hat{C}$); conditional homoskedasticity $\mathbb{E}[v_t v_t' | X_t] = \Sigma$; $\mathbb{E}[X_t X_t']$ invertible; $V = [v_1', \dots, v_m']'$ invertible almost surely; for each contrast a considered, $a' \Omega_B a > 0$ a.s. (non-degeneracy of the mixed-normal variance). $\square$